

Oil Price Shocks and India's Terms of Trade

An Empirical Assessment

Rehan Chouhan · Gaurav Arora · Munna Meetha · Sanjana

October 2026

International Economics · Group project

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P1 (15 s). Good morning. Our project asks how oil price shocks affect India's terms of trade, the price of what India sells relative to what it buys. I will cover the motivation and theory, P2 the literature and data, P3 the method and our first result, and P4 the remaining results and the policy lessons.

March 2026: India's oil shock, again

+46%

Brent crude, Feb to Mar 2026 (\$71 to \$104 a barrel)

\$113

India's crude basket in March, up from \$69 in February

Rs 94.65

per US dollar at end-March 2026, the weakest year-end rate on record

—24⁰%

India's merchandise terms of trade after the last shock, FY2020-21 to FY2022-23

Same pattern in 1973, 1979, 1990 (just before the 1991 BoP crisis) and 2022.

How much do oil shocks move India's terms of trade, and does it matter why oil became expensive?

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P1 (1 min). In March 2026 the Iran war nearly closed the Strait of Hormuz; the IEA called it the largest oil supply disruption on record. Brent rose 46 percent in a month, India's own crude basket went from 69 to 113 dollars, and the rupee ended March at 94.65 per dollar, its weakest year-end rate on record. The last time this happened, in 2022, India's merchandise terms of trade fell 24 percent in two years. This has happened before: 1973, 1979, the 1990 Gulf war just before our 1991 balance-of-payments crisis, and 2022. India buys almost all of its crude abroad, so oil is the single biggest price in its import basket. Our question: how much do these shocks move India's terms of trade, and does the cause matter?

Question and hypotheses

What we measure

Net barter terms of trade (L21)

$$\frac{\text{export price index}}{\text{import price index}} \times 100$$

Higher means each unit of exports buys more imports.

India imports about 90% of the crude it refines (RBI Table 32) and is the world's 3rd-largest oil consumer (5.4% of demand, EIA 2024).

H1 Oil price rises worsen India's terms of trade

Offer curves (L22): the oil exporter restricts supply, India's ToT ray flattens

H2 Rises and falls have unequal effects

OPEC+ market power (L16) defends price floors; Mork (1989) asymmetry

H3 Supply-driven rises hurt more than demand-driven

Offer curves, case B: a world boom also lifts demand for India's exports

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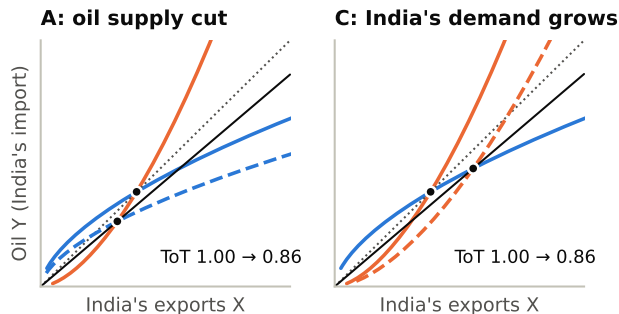
OPEC+ market power (L16) defends price floors; Mork (1989) asymmetry

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Offer curves, case B: a world boom also lifts demand for India's exports

P1 (1 min). We measure the net barter terms of trade exactly as in lecture 21: the export unit value index over the import unit value index. If it goes up, each unit of exports buys more imports. About 90 percent of the crude India refines is imported, and India is the third-largest oil consumer in the world. Our hypotheses come from the course. H1: an oil price rise worsens India's terms of trade. H2: rises and falls have unequal effects, because a cartel like OPEC+ defends price floors, as in the monopoly model of lecture 16. H3: supply-driven rises hurt more than demand-driven ones, because in a world boom other countries also buy more from India.

Theory in action: offer curves for India



India's ToT is the slope of the ray through the equilibrium (P_X/P_Y). A flatter ray means each unit of India's exports buys less oil. Data: after the 2022 shock, India's merchandise ToT fell 24% (FY2020-21 to FY2022-23).

A supply cut works like a tariff

OPEC+ offers less oil per unit of India's exports: its ToT improves, India's falls (L22).

India is not a price-taker

As the 3rd-largest buyer, India's own demand growth also worsens its ToT (L11).

World booms shift both curves

India's export demand rises too, so the sign is an empirical question: H3.

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World booms shift both curves

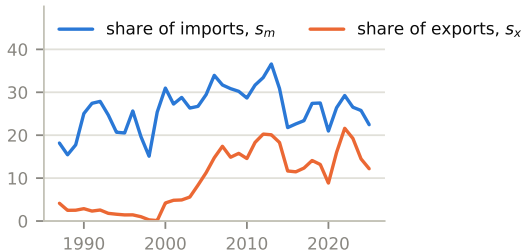
India's export demand rises too, so the sign is an empirical question: H3.

P1 (1.5 min). This is lecture 22 applied to India. India exports a composite good X and imports oil Y; each offer curve bends towards its import axis, and India's terms of trade is the slope of the ray through the crossing point. Left: an OPEC+ cut or a war means the rest of the world offers less oil for each unit of our exports. It is the lecture's tariff case with the oil exporter doing the restricting, so its terms of trade improve and India's ray flattens, from 1.00 to 0.86 in this picture. The data agree: after the 2022 shock our merchandise terms of trade fell 24 percent. Right: India is the third-largest oil consumer, so when our own demand grows our curve shifts out and our terms of trade worsen. That is why we cannot treat India as a pure price-taker, and why we later use shocks India cannot cause. A world boom shifts both curves, so theory alone cannot sign it. That is H3. Over to P2.

How big should the hit be?

$$d \ln ToT \approx (s_x - s_m) \cdot d \ln P_{oil}$$
 oil's share in exports minus its share in imports

Oil in India's merchandise trade (%)



−0.25 → −0.10

Benchmark $s_x - s_m$, FY1999-00 to FY2025-26: net oil exposure more than halved

0.06 → 4.37

Revealed comparative advantage in refined petroleum, 1999 to 2025 (Balassa, LO8)

0.55 vs 0.10

Grubel-Lloyd for oil, aggregated vs crude and products separately (L17, L19)

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P2 (1 min). Before any regression, the definition of the terms of trade already tells us roughly how big the effect should be: oil's share in exports minus its share in imports, times the change in the oil price. Oil is about a quarter of India's imports, but since the Jamnagar refineries around 2000 it has also become between 9 and 22 percent of our exports. So the predicted elasticity has gone from minus 0.25 to about minus 0.10. Two lecture measures explain why. India's revealed comparative advantage in refined petroleum rose from 0.06 in 1999 to above 4: an acquired advantage built on refining scale. And the Grubel-Lloyd index of oil trade is 0.55 when crude and products are lumped together but only 0.10 when they are separated: we import crude and export fuels, which is vertical, value-adding trade. That is India's natural hedge.

What we know, and the gap

Why ToT matter

Harberger (1950);
Laursen & Metzler
(1950): a worse ToT
lowers real income and
the trade balance.

How much is debated:
Mendoza (1995) vs
Schmitt-Grohé & Uribe
(2018).

Oil drives ToT

Backus & Crucini
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much of the variation
in ToT.

Kilian, Rebucci &
Spatafora (2009): the
effect depends on the
kind of oil shock.

Shocks differ

Hamilton (1983, 2003);
Mork (1989):
asymmetry.

Kilian (2009);
Baumeister & Hamilton
(2019); Känzig (2021):
supply vs demand.

India

Bhanumurthy, Das &
Bose (2012); Ghosh
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Focus: trade balance,
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Gap. Indian studies model the trade balance, inflation or the rupee. We found none that models India's terms of trade directly, with a theory benchmark, identified supply and demand shocks, and the 2022 and 2026 episodes.

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P2 (1 min). Four strands. Why the terms of trade matter: Harberger and Laursen-Metzler show a worse terms of trade lowers real income and the trade balance; how much they drive business cycles is debated, Mendoza says a lot, Schmitt-Grohé and Uribe find under 10 percent. Our anchor paper is Backus and Crucini (2000): oil explains much of the terms-of-trade variation of industrial countries. Kilian, Rebucci and Spatafora show the external-balance effect depends on the kind of oil shock. On defining shocks: Hamilton and Mork on asymmetry, and Kilian, Baumeister-Hamilton and Känzig on separating supply from demand. Indian work looks at the trade balance, inflation and the rupee. So our gap is the terms of trade itself, with a theory benchmark, identified shocks and the latest episodes.

Data: official sources, checked line by line

Official agencies and published replication data only

RBI Handbook 2025-26	UNCTADstat
DGCI&S trade indices	UN Comtrade
PPAC (Indian basket)	US EIA BIS
World Bank Pink Sheet	Baumeister-Hamilton
World Bank WDI	Känzig (AER data)

Monthly: Apr 2019–Jun 2026 (87 months)

Annual: FY1970-71 to FY2025-26

Merchandise ToT: FY1994-95 to FY2025-26

o failed checks

recomputed indices, cross-source comparisons,
fiscal-year alignment

Problems we found in official tables

- RBI's base-year ToT series contradict each other: +21.7% vs -21.0% (1999–2007)
- RBI Table 32 repeats the 2000–07 rows for 1990–97
- DGCI&S export volume index jumps between 54 and 2001, so we use price indices only
- Three published RBI ToT values disagree with RBI's own indices

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P2 (1.5 min). Our data come only from official agencies and from the published data of peer-reviewed papers: the RBI Handbook, DGCI&S trade indices, PPAC, the World Bank, UNCTAD, UN Comtrade, the US EIA, the BIS, and the Baumeister-Hamilton and Känzig shock series. Every file is logged with its web address and a checksum, and a validation script runs all the cross-checks: none fails. Checking mattered. RBI's three base-year series of the terms of trade move in opposite directions over 1999 to 2007; RBI's crude-import table repeats the 2000-07 rows for 1990-97; the DGCI&S export volume index jumps between 54 and 2001; and three published RBI values disagree with RBI's own indices. We documented and corrected each one. Over to P3.

Method in four steps

1

Unit-root tests

ADF, Phillips–Perron,
KPSS, Zivot–Andrews

All series stationary
in differences, none
 $I(2)$, so ARDL is valid

2

ARDL bounds test

H1: short-run vs
long-run effect,
speed of adjustment

Works with mixed
 $I(0)/I(1)$ and 54 years;
short vs long run as
in L13

3

Nonlinear ARDL

H2: separate effects
of oil price rises and
falls

Wald tests of
long-run and
short-run symmetry

4

Shock split + local projections

H3: supply- vs
demand-driven oil
prices; monthly
timing

Baumeister–
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Wald tests of long-run and short-run symmetry

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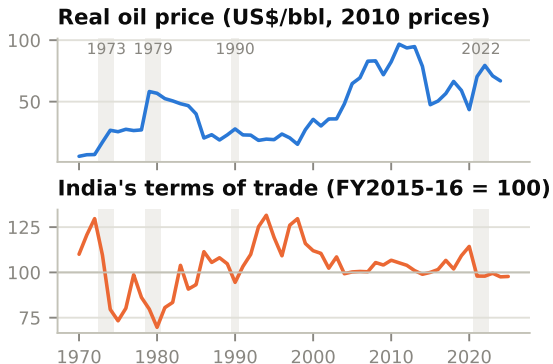
Shock split + local projections

H3: supply- vs demand-driven oil prices; monthly timing

Baumeister-Hamilton and Känzig shocks are not caused by India

P3 (1 min). Four steps. First, unit-root tests, including Zivot-Andrews with a break. Every series is stationary in first differences and none is $I(2)$, so ARDL is valid. Second, the ARDL bounds test of Pesaran, Shin and Smith for H1. It handles a mix of $I(0)$ and $I(1)$ variables, works with our 54 years, and separates the short run from the long run, like the time horizons of lecture 13. We use finite-sample critical values. Third, the nonlinear ARDL, which splits oil prices into rises and falls, for H2. Fourth, for H3 we split each oil price change into a supply-driven and a demand-driven part using Baumeister-Hamilton's shocks, and we trace monthly timing with local projections. These shocks come from the world oil market and OPEC announcements, so India's own demand cannot cause them.

H1: oil hurts India's terms of trade, and fast



—0.24

Short-run elasticity: a 10% oil rise cuts ToT by 2.4% the same year ($p < 0.001$, robust s.e.)

31% a year

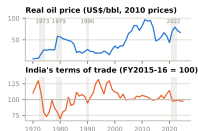
of any gap closes each year (error correction $\alpha = -0.31$): the hit is large but temporary

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Long-run elasticity, not significant; bounds $F = 4.74$ ($p = 0.07$)

—0.19 to —0.28 in six specifications; diagnostics pass; CUSUM stable.

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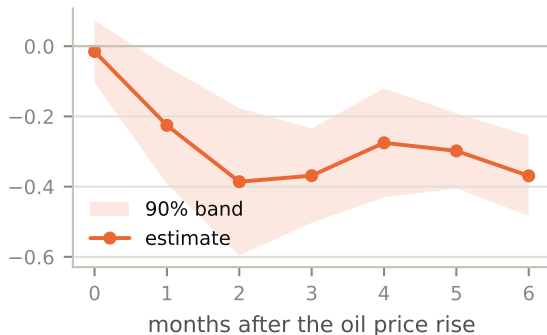
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P3 (1.5 min). The charts cover 55 years: every big oil spike, 1973, 1979, 1990, 2022, lines up with a fall in India's terms of trade. The ARDL confirms H1. The short-run elasticity is minus 0.24: a 10 percent rise in the real oil price cuts the terms of trade by 2.4 percent in the same year, significant well below 1 percent even with robust standard errors. The error-correction coefficient is minus 0.31, so about 31 percent of any deviation closes each year. The long-run elasticity is small, minus 0.09, and not significant; the bounds F of 4.74 rejects no relationship only at 10 percent against the upper bound. So oil moves our terms of trade strongly but temporarily. The short-run effect holds in all six specifications. Diagnostics: no serial correlation, normal errors, correct functional form, stable CUSUM; Breusch-Pagan found heteroskedasticity, so we report robust standard errors.

The hit arrives within two to three months

ToT response (%) to a 1% Brent rise



—0.39

Elasticity two months after an oil rise; almost nothing in month 0 (monthly data, Apr 2019–Jun 2026)

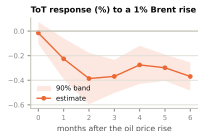
1.70

Benchmark test: theory predicts a coefficient of 1 on $(s_x - s_m) \Delta \ln P_{oil}$; 1 is not rejected at 5% ($p = 0.07$)

+18% vs +16%

2026 so far: export prices rose as fast as import prices (Feb–Apr), so ToT has not clearly fallen yet

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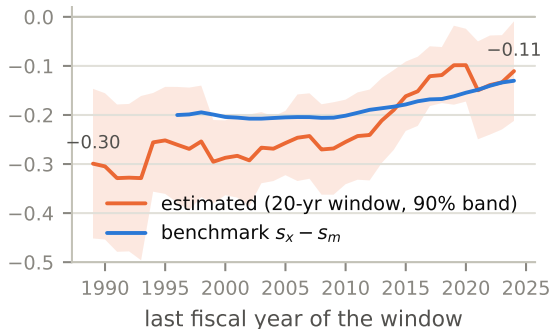
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2026 so far: export prices rose as fast as import prices (Feb–Apr), so ToT has not clearly fallen yet

P3 (1.5 min). How fast does it happen? With monthly DGC&S data we run local projections. In the month of the price change almost nothing happens; after one to three months the terms of trade fall by about 0.2 to 0.4 percent per 1 percent oil rise, and the effect stays. The lag fits how crude is bought: cargoes are priced on earlier benchmarks and recorded at customs when they arrive. Is the size what theory predicts? Our benchmark test regresses the change in the terms of trade on the share-weighted oil change; theory says the coefficient is 1. We get 1.70 and cannot reject 1 at 5 percent. The extra comes from indirect channels like fertiliser, petrochemicals and freight. And 2026 shows the hedge working: from February to April our export unit values rose 18 percent, as fast as import unit values, so up to June the terms of trade had not clearly fallen. Single months are noisy, so we judge episodes on averages, never on one month. P4 now covers H2, H3 and policy.

Symmetric, and half as large as before

Oil elasticity of India's ToT



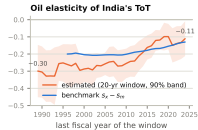
H2 not supported

Short run: rise -0.27 , fall -0.22 ; symmetry never rejected ($p = 0.54-0.91$). A long-run asymmetry appears only with the 1970s in the sample ($p = 0.28$ from 1980).

Exposure halved

Elasticity -0.30 (window to 1989) to -0.11 (to 2024). It tracks the benchmark ($-0.25 \rightarrow -0.10$) as refined-fuel exports grew: the refining advantage works as a hedge ($RCA \approx 4$).

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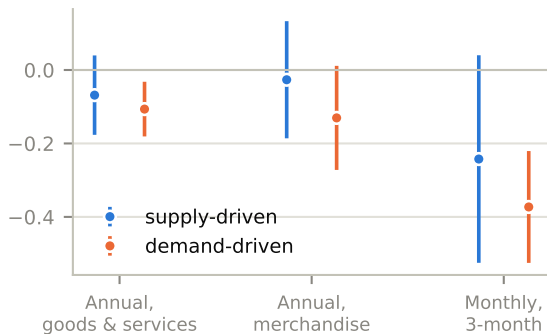
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P4 (1.2 min). H2 asked whether rises and falls have unequal effects. In the short run, clearly not: minus 0.27 for a rise, minus 0.22 for a fall, and symmetry is never rejected. The baseline long-run model finds a slightly bigger response to falls, but it disappears once we drop the 1970s and does not hold for merchandise trade. So we call the effects symmetric, and H2 is not supported: India gains from cheaper oil about as much as it loses from dearer oil. The chart shows a second and robust finding: India has become much less exposed. A 10 percent oil rise used to cut the terms of trade by about 3 percent; now it is about 1.1 percent. That matches the benchmark from P2's slide and the rise of refined-fuel exports.

H3: how much oil rises matters more than why

ToT effect of a 1% oil rise, by cause



H3 not supported

Demand-driven rises hurt at least as much as supply-driven ones ($p = 0.53$ – 0.71 for a difference).

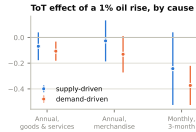
Why

A world boom also lifts the prices of gold, coal, fertiliser, edible oil and metals that India imports.

Cross-check

Känzig's OPEC-news shocks (about +10% on oil): India's ToT falls 4–5% two to three months later.

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Känzig's OPEC-news shocks (about +10% on oil): India's ToT falls 4-5% two to three months later.

P4 (1.2 min). H3 asked whether the cause matters. We split every monthly oil price change since 1975 into a supply-driven and a demand-driven part using Baumeister and Hamilton's shocks, then asked which part moves India's terms of trade. In all three samples the demand-driven part hurts at least as much, and the difference is not significant, so H3 is not supported. The offer-curve idea that a boom also raises demand for our exports does not show up, because a global boom raises the prices of everything India imports, gold, coal, fertiliser, edible oils and metals, while our manufactured exports gain less. Känzig's OPEC announcement shocks confirm that pure supply shocks do hurt: about 4 to 5 percent after two to three months. For India, how much oil rises matters more than why.

What it means for India

What we found

- H1** A 10% oil rise cuts India's ToT by 2.4% within a year, mostly within 2–3 months
- H2** Rises and falls matter about equally
- H3** Demand-driven rises hurt as much as supply-driven ones
 - + Exposure has halved since the 1990s: refining is India's hedge

What India can do

Buffers for both directions: strategic reserves and a fuel-tax buffer that cuts duties in spikes and rebuilds them in slumps

Grow the refining hedge: more refined and petrochemical exports raise s_x (L08, L16)

Diversify suppliers: more sources weaken any one seller's market power (L16)

Cut commodity dependence: energy transition and ethanol blending, since demand booms hurt too

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Diversify suppliers: more sources weaken any one seller's market power (LsB)

Cut commodity dependence: energy transition and ethanol blending, since demand booms hurt too

P4 (1.1 min). To sum up. H1 holds: a 10 percent oil rise cuts India's terms of trade by about 2.4 percent within a year, mostly within two to three months. H2 and H3 do not: rises and falls matter about equally, and demand-driven rises hurt as much as supply-driven ones. And our exposure has roughly halved since the 1990s because refining became an export strength. The policy lessons: because the effects are symmetric, India needs buffers that work both ways, like strategic reserves and a fuel-tax buffer; keep growing the refining hedge; diversify suppliers; and since demand booms hurt too, cut commodity-import dependence through the energy transition. Thank you, we are happy to take questions.

Thank you

Questions welcome

Backup: oil episodes, fiscal-year averages

Episode	Fiscal years	Real oil	ToT (G&S)	ToT (merch.)
1973 OPEC embargo	1972-73 to 1974-75	+284.1%	−38.6%	−
1979 Iranian revolution	1978-79 to 1980-81	+110.2%	−19.2%	−
1990 Gulf war	1989-90 to 1990-91	+21.1%	−9.8%	−
2008 price spike	2006-07 to 2008-09	+19.9%	+4.9%	+5.8%
2014-16 price collapse	2013-14 to 2015-16	−49.9%	−1.0%	+19.4%
2020 COVID crash	2019-20 to 2020-21	−26.4%	+4.7%	+9.6%
2022 Russia-Ukraine	2020-21 to 2022-23	+82.8%	−14.4%	−23.7%

2008 looks mild in annual averages because the spike lasted only months; the 2014–16 collapse shows the merchandise gain from cheap oil.

Backup: unit-root tests (p-values)

Series	ADF lvl	ADF diff	PP lvl	PP diff	KPSS lvl	ZA lvl	ZA break
ln ToT goods & services	0.057	0.000	0.050	0.000	0.100	0.391	1985
ln ToT merchandise (FY1994-)	0.271	0.001	0.392	0.000	0.010	0.062	2011
ln real oil price	0.053	0.000	0.050	0.000	0.013	0.533	1999
ln real non-energy prices	0.355	0.000	0.395	0.000	0.100	0.911	2005
ln real gold price	0.945	0.000	0.582	0.000	0.010	0.942	2006

ADF and PP: null of a unit root. KPSS: null of stationarity. ZA: unit root with one break. Every series is stationary in differences, so none is $I(2)$.

Backup: ARDL main estimates

	Estimate	Std. error	p-value
Short-run oil elasticity (same year)	−0.241	0.049	0.000
Long-run oil elasticity	−0.092	0.059	0.119
Long-run non-oil commodity elasticity	−0.249	0.177	0.159
Error-correction speed α	−0.308	0.086	0.001

Bounds $F = 4.74$, $p = 0.025$ against $I(0)$ and 0.070 against $I(1)$. Diagnostics (p): Breusch–Godfrey 0.90 , Breusch–Pagan 0.019 (so HC1 robust s.e.: 0.040 for the short-run effect), Jarque–Bera 0.25 , RESET 0.84 ; CUSUM and CUSUMSQ inside 5% bands.

Backup: ARDL robustness

Specification	N	SR oil	t	LR oil	ECT α	F	p (I(1))
Baseline: oil + non-oil (G&S ToT)	54	−0.241	−4.931	−0.092	−0.308	4.743	0.070
Oil only	54	−0.195	−4.577	−0.084	−0.241	4.064	0.173
Oil + non-oil + gold	54	−0.217	−4.738	−0.061	−0.310	3.538	0.149
Brent instead of average crude	54	−0.236	−4.972	−0.089	−0.307	4.709	0.072
Post-1980 sample	44	−0.211	−4.965	−0.192	−0.355	3.651	0.172
Merchandise ToT, FY1994-2024	30	−0.277	−3.457	−0.146	−0.368	2.084	0.506

Backup: NARDL symmetry tests

Specification	N	L^+	L^-	p (LR)	π^+	π^-	p (SR)
Baseline FY1970-2024, with non-oil control	54	-0.162	-0.228	0.035	-0.271	-0.215	0.620
No control	54	-0.199	-0.285	0.017	-0.233	-0.192	0.735
Brent	54	-0.157	-0.221	0.036	-0.261	-0.217	0.693
Extra lag of ΔToT	53	-0.119	-0.180	0.051	-0.239	-0.215	0.831
FY1975-2024	49	-0.106	-0.161	0.077	-0.217	-0.204	0.913
FY1980-2024 (drops the 1970s shocks)	44	-0.114	-0.145	0.275	-0.245	-0.172	0.537
Merchandise ToT FY1994-2024	30	-0.187	-0.075	0.131	-0.234	-0.308	0.746

$L^\pm$: long-run effects of rises and falls; $\pi^\pm$: short-run. Short-run symmetry is never rejected; the long-run asymmetry needs the 1970s.

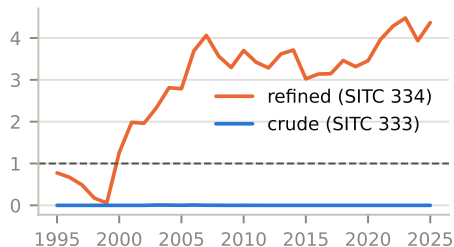
Backup: supply- vs demand-driven oil prices

Sample	N	Supply	s.e.	Demand	s.e.	p (equal)
Annual G&S ToT, FY1975-2025	51	−0.068	0.066	−0.106	0.045	0.707
Annual merchandise ToT, FY1995-2024	31	−0.027	0.097	−0.130	0.086	0.547
Monthly merchandise ToT, 3-month cumulative, 2019-26	82	−0.242	0.172	−0.373	0.093	0.535

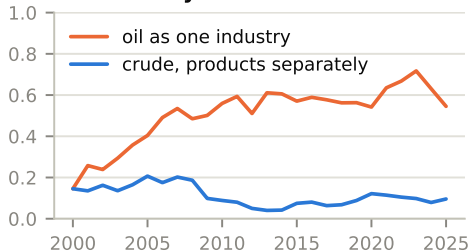
Supply part = fitted contribution of Baumeister–Hamilton supply shocks to the oil price change;
demand part = activity, consumption and inventory shocks. Känzig news shock: annual −0.95% per shock (s.e. 0.49); monthly −4.2% and −4.7% at 2 and 3 months.

Backup: the refining hedge in trade-theory terms

Revealed comparative advantage



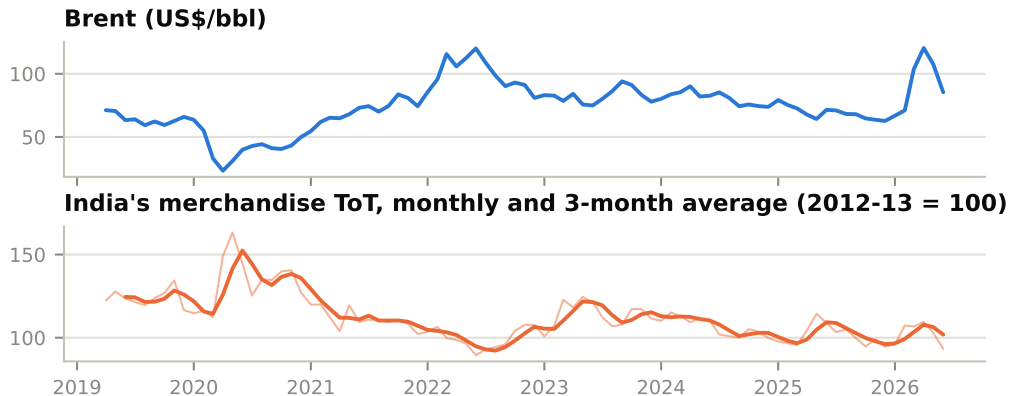
Grubel-Lloyd index of oil trade



Refined exports are worth 1.13–1.41 times crude imports per tonne (2000–2025), outside the $\pm 15\%$ band that L2O uses for horizontal trade in every year except 2011 and 2012. So this is vertical, value-adding trade: India imports crude and sells refined fuel.

Year	Crude import (US\$/t)	Refined export (US\$/t)	Ratio
2005	361.40	462.95	1.28
2015	368.81	520.32	1.41
2022	713.15	996.29	1.40
2024	585.45	721.25	1.23
2025	528.03	672.85	1.27

Backup: monthly terms of trade, 2019–2026



Backup: limitations and the data fixes we made

- Unit value indices are not pure price indices: they mix composition and quality changes
- Monthly DGCI&S data start in April 2019; earlier monthly series are not published online
- DGCI&S's two monthly bases (2012-13 and 2022-23) disagree month to month, so no episode is judged on a single month
- The goods-and-services ToT includes services (software), which dilutes the oil effect
- Imports are valued CIF and exports FOB, so freight and insurance spikes also move measured ToT (L14)
- Fixes: RBI base-year series linked at overlaps; NTT recomputed from the price indices; duplicated RBI rows dropped; WDI 1960s deflators excluded; mis-served PPAC file set aside
- The 2026 Hormuz shock is still unfolding (ToT data to June 2026)